Saransh Shankar

Software Engineer — Backend | Infrastructure | Distributed Systems
saransh.shankar@gmail.com | +91 7406320038 | India | linkedin.com/in/saransh-shankar-93ab65223
github.com/Wise-Wizard | codeforces.com/profile/Wise_Wizard | leetcode.com/Wise-Wizard

Education

Indian Institute of Information Technology, Gwalior — Gwalior, India
Integrated B.Tech + M.Tech in Information Technology — *Sept 2021 – May 2026*

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming

Experience

Backend Engineering Intern
Origin — autonomous construction robotics (ROS 2, NVIDIA Orin) — *Jan 2026 – Present*

- Enabled **pre-merge regression testing against real robot data** by designing and building, as **sole owner**, a module-wise **rosbag replay framework** (Python, Docker, ROS 2): replays recorded field data through any code version of any of 6 autonomy modules in isolation and validates outputs against per-module criteria in **CI** — field failures that previously needed a robot now reproduce on a laptop; delivered end-to-end from proposal and **system design (HLD/LLD)** to implementation.
- Ensured **no field incident is ever lost** by building an always-on **C++ flight recorder**: a disk-segmented MCAP ring buffer over 60+ debug topics with **O(1) hardlink-based incident preservation** (no copy, no write stall) — validated on robot hardware at **28.65 MB/s sustained within 71 MB RAM**.
- Cut fleet debugging from **SSH-into-the-robot to searchable-in-seconds** by building the end-to-end **observability pipeline**: structured async-logging libraries for ROS 2 (C++/Python), loss-tolerant cross-device log transport (**rsyslog/RELP** with disk-buffered queues that survive network partitions), and **Vector, Loki, Prometheus, Grafana** dashboards in production.
- Freed **70% of compute on NVIDIA Orin** for perception and planning by migrating backend services to the V2 architecture — refactored hot-path modules and eliminated per-request subprocess overhead.
- Standardized **what every robot records** by building a task-scoped **C++ rosbag recording service** and owning the cross-team 64-topic recording catalog — the data foundation the replay framework runs on.

Software Engineer
Camb AI — voice and speech AI — *May 2025 – Aug 2025*

- Made Camb's speech AI reachable from **every major agent ecosystem** by independently shipping **10+ integrations** (LiveKit, Pipecat, LangChain, CrewAI, LlamaIndex), a **production MCP (Model Context Protocol) server**, and an **n8n community node published on npm** — each owned from prototype to release.
- Enabled community-driven speech-dataset creation for the low-resource **Maleku language** by building a **WhatsApp TTS data-collection platform** automating queued prompts and recording collection.
- Cut audiobook production **from days to minutes** with an automation pipeline (OCR + Gemini + Camb AI TTS) converting raw PDFs into finished audiobooks.
- Cut microservice **CI build time 67%** with a cached base image and a layered Dockerfile strategy.

Software Engineering Intern (LFX Mentorship)
Jaeger Tracing — CNCF open source — *June 2024 – Sept 2024*

- Engineered the **OpenTelemetry metrics adapter for Jaeger V2**, improving metrics-collection speed **40%** and establishing **observability parity between Jaeger V1 and V2** via a unified telemetry container.
- Streamlined the OTEL Collector (**TracerProvider/MeterProvider**) to remove redundant internal telemetry and integrated the **health-check extension** for endpoint liveness monitoring across V2 components.

Earlier Internships: Data Scientist, Credohire (Sept–Nov 2023) — NLP-based candidate assessment; Full-Stack Developer, Andwemet (May–Sept 2023) — React platform serving 10,000+ users; Software Developer, Epvi India (Oct 2022–Jan 2023) — Flutter app shipped with 95% test coverage.

Projects

Deep-Learning Intrusion Detection System — *Peer-reviewed publication, IEEE Xplore*

- Published peer-reviewed research on a network **intrusion detection system**: deep learning on the CIC-IDS dataset (**100,000+ samples**) with **DSSTE** to correct class imbalance, achieving **99.5% accuracy** on threat classification.

Technical Skills

Languages: Python, C++, Go, SQL, JavaScript/TypeScript, Bash
Backend & Distributed Systems: REST APIs, Microservices, FastAPI, PostgreSQL, Redis, Celery, MongoDB, Node.js, System Design
Infrastructure & DevOps: Linux, Docker, Kubernetes, CI/CD (GitHub Actions), AWS (S3, ECS), Git, ROS 2 (rclcpp/rclpy)
Observability & Monitoring: OpenTelemetry, Jaeger, Prometheus, Grafana, Loki, Vector, rsyslog
Testing & ML: pytest, unit/integration testing, TensorFlow, Scikit-Learn, Pandas, MCP, LangChain

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | OCI 2025 AI Foundations Associate**